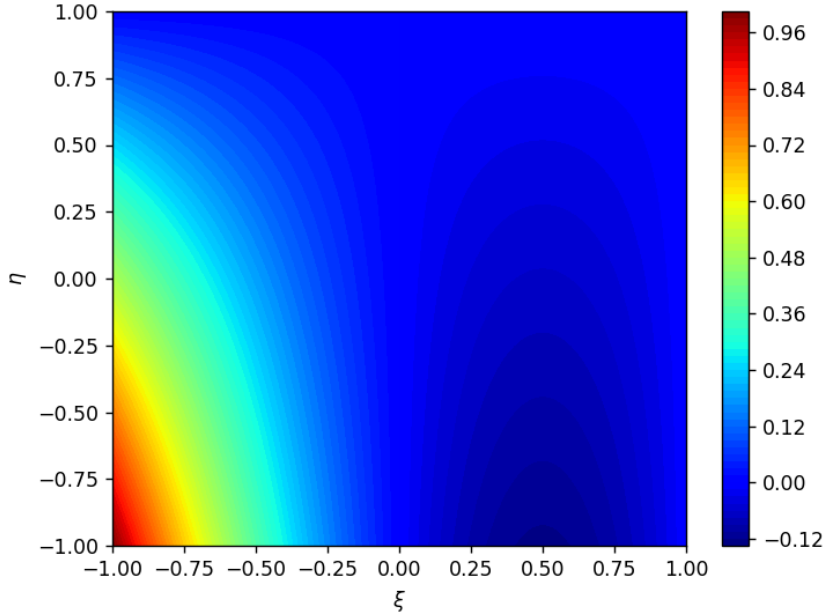
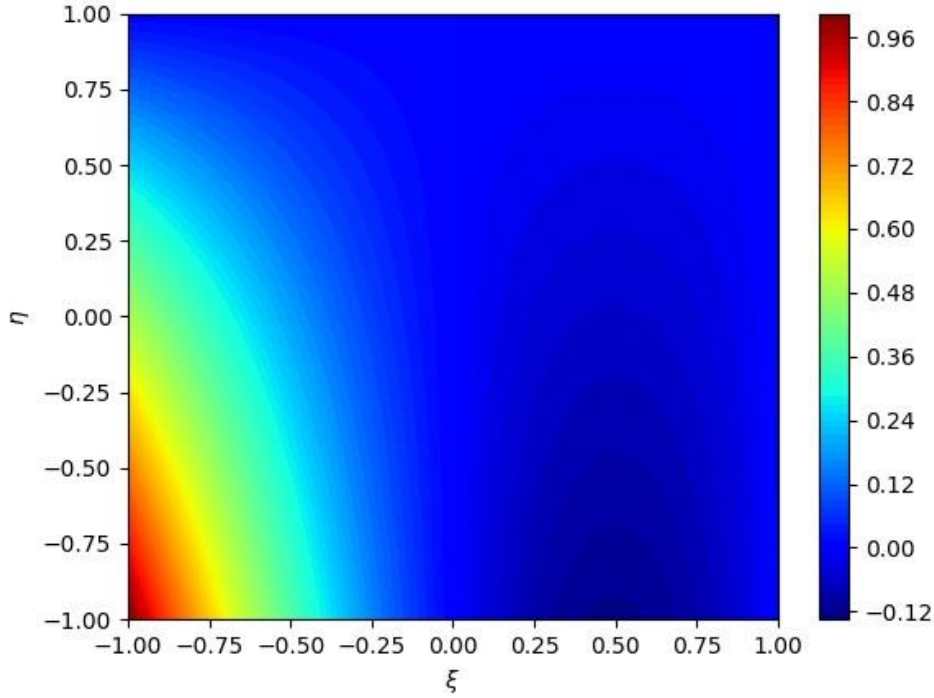
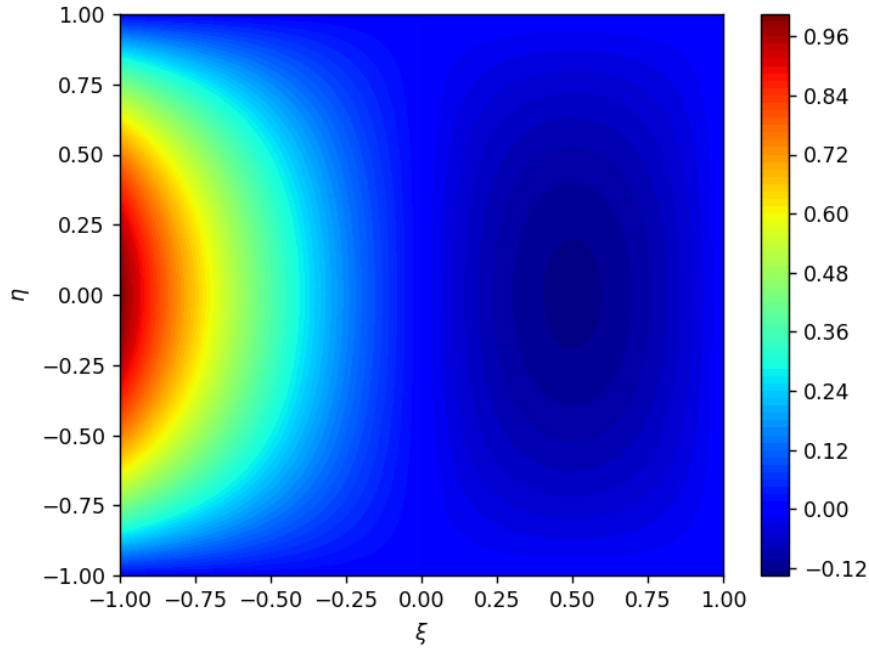
[1, 1]


1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
η
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
ξ
0.96
0.84
0.72
0.60
0.48
0.36
0.24
0.12
0.00
−0.12

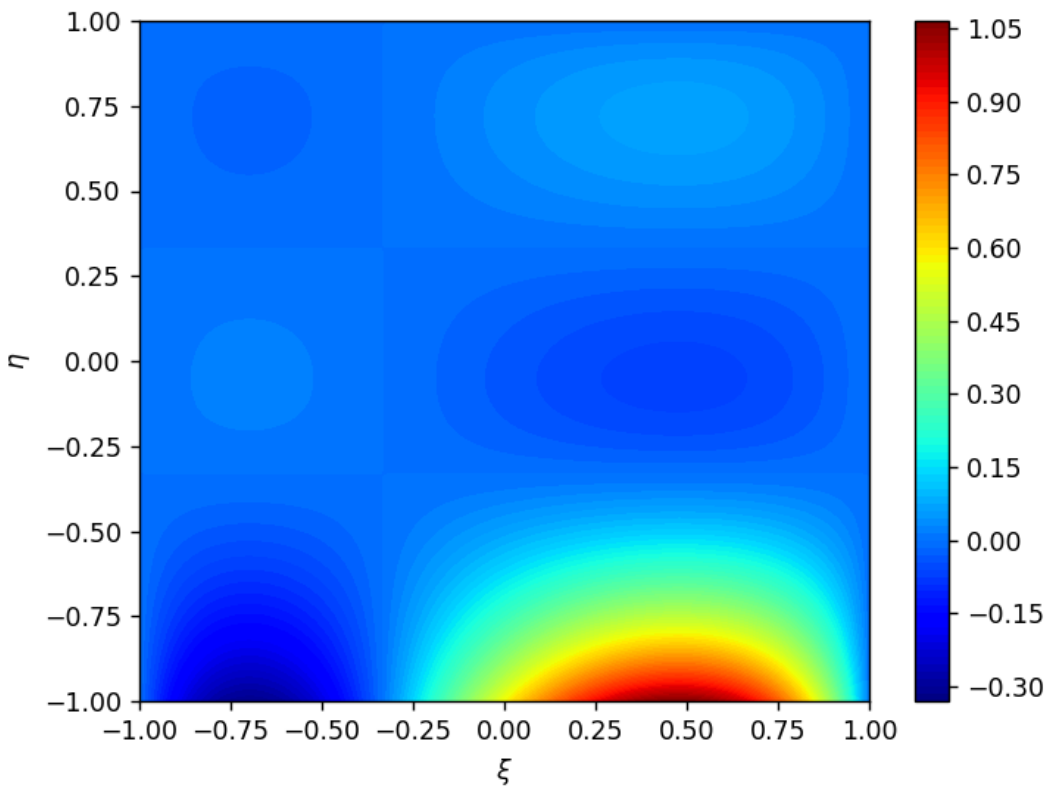
[2, 1]


1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
η
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
ξ
0.96
0.84
0.72
0.60
0.48
0.36
0.24
0.12
0.00
−0.12

[2, 2]

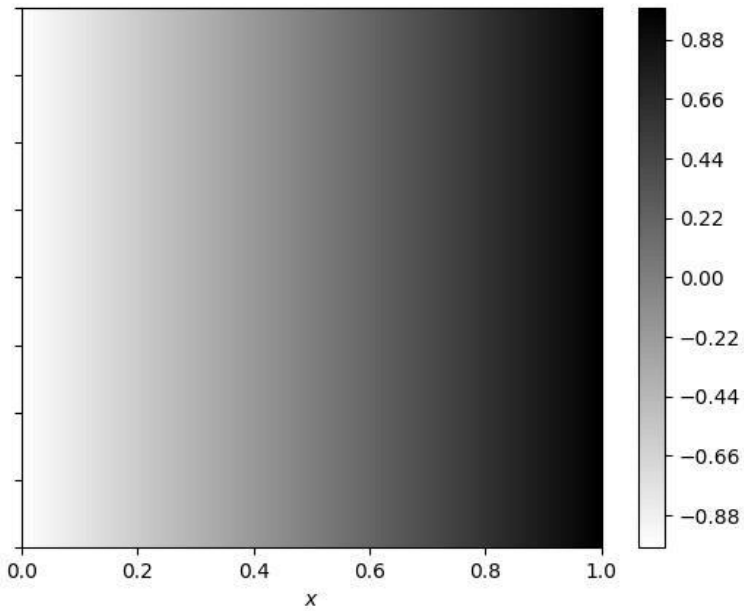

1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
η
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
ξ
0.96
0.84
0.72
0.60
0.48
0.36
0.24
0.12
0.00
−0.12

[3, 3]

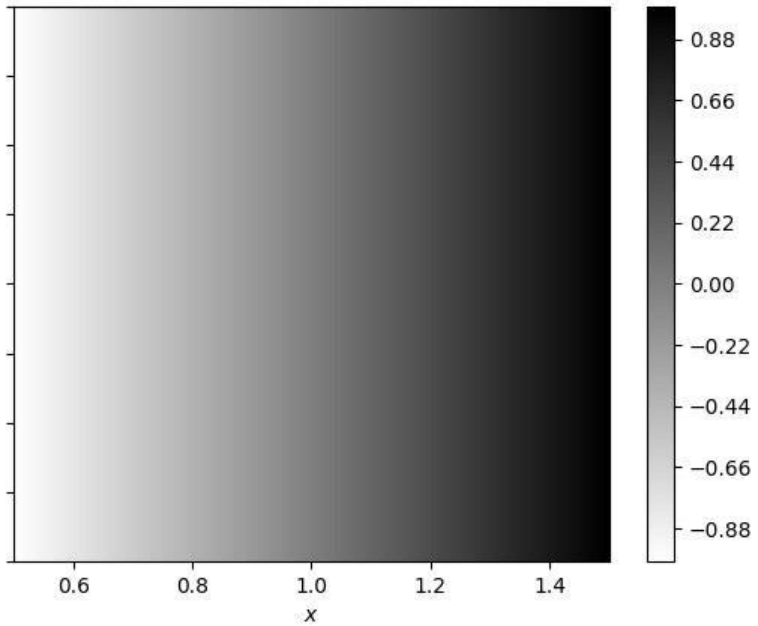

1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
η
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
ξ
1.05
0.90
0.75
0.60
0.45
0.30
0.15
0.00
−0.15
−0.30

XMAP:

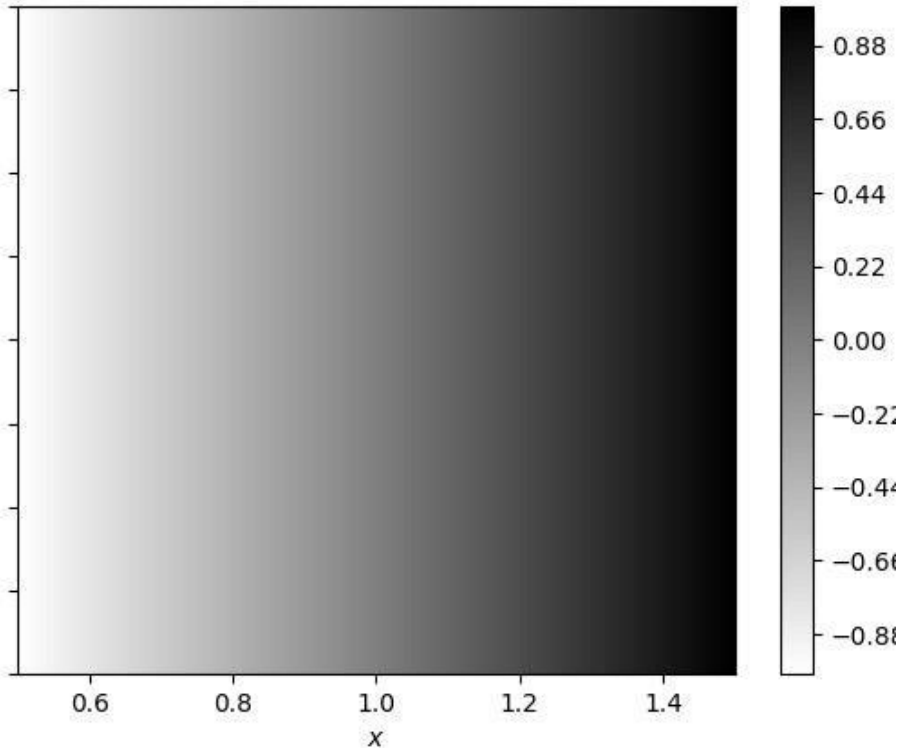
Case 1:



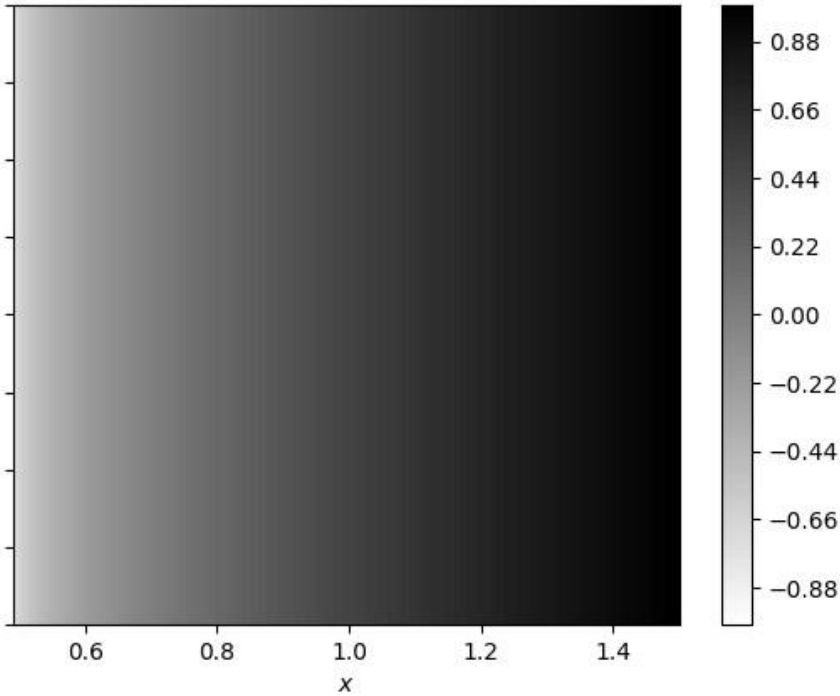
Case 2:



Case 3:


0.88
0.66
0.44
0.22
0.00
−0.22
−0.44
−0.66
−0.88
0.6
0.8
1.0
1.2
1.4
x

Case 4:


0.88
0.66
0.44
0.22
0.00
−0.22
−0.44
−0.66
−0.88
0.6
0.8
1.0
1.2
1.4
x

When is the mapping xe( ) linear? When is this map transition from the interval [-1, 1] to the input spatial element in a manner that correctly transforms points for Gaussian quadrature?

Cases 1-3 are linear. These are the cases that allow for the proper transformation for Gaussian quadrature, the mapping must be linear.